John Hua Yao

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PUBLICATIONS

John Hua Yao, Qi Wu, Yihuai Gao, Chelsea Finn, Zipeng Fu. ALOHA Lightning: Learning Fast and Precise Manipulation**.** *in submission International Conference on Robotics and Automation (ICRA) 2026.*

Jorge Carrasco Pollo†, Ioannis Kapetangeorgis†, Joshua Rosenthal†, **John Hua Yao**†. [Re] Benchmarking LLM Capabilities in Negotiation through Scoreable Games**.** *Transactions on Machine Learning Research (TMLR) 2025.*

EDUCATION

University of Amsterdam (Amsterdam, Netherlands) *September 2024 - September 2026*
- Master's of Science in Artificial Intelligence. Current **9.0** GPA (**4.0** US, top 2.7%). *Relevant taught topics: diffusion, flow matching, geometric machine learning (steerable CNNs, conditional neural fields), sparse attention, spatial LLM..*

Imperial College London (London, UK) *October 2018 - October 2022*
- (Integrated) Master's of Engineering in Mathematics and Computing. Received 2:1 w/Honors.

RESEARCH

IRIS Lab, Stanford University *July 2025 - Present*
Visiting Researcher. Advisor: Zipeng Fu, Chelsea Finn
- Created a full stack system for fast trajectory collection, training, and test time optimization enabling high speed, precise manipulation policies.
- Designed the gripper lead handles, established experimental conditions and tasks, created a visual masking mechanism and an inference time smoothing algorithm, and overall integrated various hardware components. Additionally, building SDKs for novel robotic hardware for future adaptation.

Department of Informatics, University of Amsterdam *January 2025 - March 2025*
Course Project. Advisor: Alexandre da Silva Pires
- Reproduced a paper on LLM negotiation published at NeurIPS 2024. Examined the potential usefulness of the proposed benchmarking tool on LLM capabilities, and proposed alternative metrics for greater transparency.

Department of Mathematics, Imperial College London *January 2022 - July 2022*
Master's Thesis. Advisor: Jeroen Lamb
- Constructed a novel learned aggregation architecture on graph neural networks using the fundamental theorem of symmetric polynomials. Theoretical concept for an alternative system for group invariance in learning systems.

Spaceflight Telemetry Flight Computer (Open-Source) *October 2021 - July 2022*
- Developed the main control loop and telemetry logging mechanism of the in-flight computer of 'Project Mach One', a rocket faster than the speed of sound, at Imperial Space Society.
- Implemented mission-critical actions such as parachute deployment and logging at 10Mb/s, using Arduino.

PathBench3D (Open-Source) *October 2020 - January 2021*
- Implemented 3D visualization of the open-source PathBench robotics research software in Python with plugins to ROS, adding new functionalities such as camera scrolling, transparency sliders for blocks, and algorithm visualization.

PROFESSIONAL

Dutch NAO Robotics Team *September 2024 - Present*
- Developed and prototyped reinforcement learning policies for robotic football player **high level ball search** behavior.
- Implemented an unscented Kalman filter in Rust with **Rerun** for **motion tracking** of a moving soccer ball.
- Created new leads for **fundraising** efforts by engineering a new multi-tiered sponsorship program and pitching to live audiences at fundraising events on campus. Lead many **outreach** classes for local community interest groups.

University of Amsterdam, Informatics Department Amsterdam, Netherlands
Teaching Assistant *January 2025 - March 2025*
- (1) Behavior Based Robotics. *January 2025.* Developed and debugged motion control and computer vision codebase for students to operate the Nao v6 robot. Hosted scientific writing seminars based on robotics-themed topics.
- (2) Autonomous Mobile Robotics. *February - March 2025*. Taught theory of **localization**, **kinematics** and **control**.

Angstrom Sports London, UK
Software Engineer *January 2023 - June 2024*
- Implemented **MCMC** sports simulation performance rewrite, bringing average simulation time from 1500 ms to 700 ms, halving latency for online pricing systems and leading to improved margins for downstream clients.
- Optimized thread management and processor usage, created CentOS server running profiles including **Linux** boot optimizations, and simplified monolith code structures into segmented microservices in **C#.**